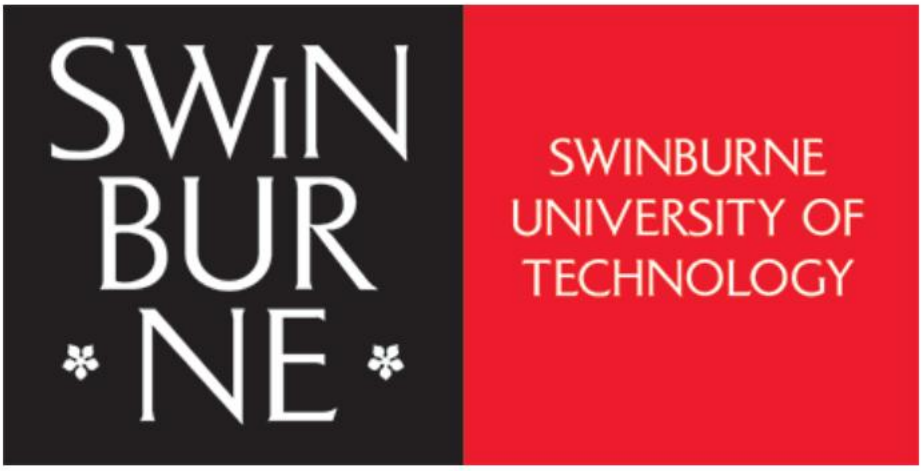


2026-COS30049-Computing Technology Innovation Project

Workshop Guide

Workshop 01
Introduction to the Unit & Project selection and initiation

Objective: By the end of this workshop, students should have a solid understanding of the course structure and have started forming teams. They should have a grasp of project management principles and be able to comprehend the project development process.

Workshop Structure:

1. Ice-breaker (20 minutes):

Each student will take 1-2 minutes to introduce themselves (could include their name, background or major, personal interests, skills, and what they hope to gain from this workshop.

2. Introduction to the unit (15 minutes):

Emphasizing the unit structure, assessments, and unit pass criteria.

3. Introduction (25 minutes):

Recall the concepts we learned in Week1:
a. Artificial Intelligence
b. Web Development
c. Project Management phases
d. Business Case
e. Strategic Plan
f. SWOT Analysis
g. Web Design: How to define requirements

4. Introduction to Our Cybersecurity Topics (15 minutes)

The topic you choose here will carry through from Assignment 1 to Assignment 3, so it can't be changed later. Please think carefully and
pick something you're genuinely interested in.

1. **Prompt Injection Guardrail (AI security)**: The user types a message intended for an AI assistant. Your app decides whether to

allow or block it, names the attack type, and highlights the words that triggered the decision. The challenge is that attackers disguise their text, so you must design signals that survive character tricks and encoding.

2. **AI-Generated Content Detection (AI governance)**: The user pastes a piece of writing. Your app scores each sentence, colours the passage by likelihood, and gives an overall estimate with a confidence range. You decide which writing-style measurements to compute and how to combine sentence scores into one document score.

3. **Phishing URL Detection (Web security)**:The user pastes a web address. Your app returns a trust score and shows which parts of the address look suspicious, such as excessive length, an IP address in place of a domain name, look-alike characters, or an unusual top-level domain. You write the code that turns a raw URL into those signals. Runs fully offline, with no page fetching.

4. **Log Anomaly Detection (Intrusion security)**: The user uploads a log file from a distributed system. Your app groups the entries into sessions, decides which sessions look abnormal, and presents the findings on a monitoring dashboard with drill-down into individual sessions. You write the code that turns raw log lines into structured events, which is the hardest part of this topic and also the most useful.

5. Group Finding (25 minutes)

Divide students into groups of 3. Once grouped, students should discuss each of the topics listed bove. Alternatively, students who already share common interests can form teams on their own during the same session. Students are also encouraged to propose a topic of their own interest for analysis. However, they must demonstrate that further investigation has been conducted and seek approval from their session tutor. During discussion, students should consider:

- The potential benefits each project could bring
- The challenges they might face
- The societal impact their proposed solutions may have

6. Lastly, let students select groups (5 mins)

Step 1:
Go to the People section and select a group set.
Step 2:
To join a group from the list, simply select the one th at best aligns with your team and click to confirm your choice. Feel free to switch between different groups until you find the right fit. However, please note that once you have submitted the first assignment, group changes will no longer be allowed.

7. Reflection and Closing (5 minutes):

Allow students to share what they learned, found interesting, or had difficulty understanding.